

Structural Analysis of the EGFR-Gefitinib Complex Using PyMOL

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1. Introduction

The Epidermal Growth Factor Receptor (EGFR) is a **receptor tyrosine kinase** that plays an important role in regulating cell growth, proliferation, and survival. Abnormal activation or overexpression of EGFR has been associated with several types of cancer, including **non-small cell lung cancer (NSCLC)**.

Gefitinib is a small-molecule **tyrosine kinase inhibitor** that specifically targets the ATP-binding site of EGFR. By binding to the kinase domain, Gefitinib prevents ATP binding and inhibits downstream signaling pathways involved in tumor growth.

Structural visualization of protein–ligand complexes provides valuable insights into molecular recognition and drug-target interactions. In this project, the crystal structure of the **EGFR kinase domain bound to Gefitinib (PDB ID: 4WKQ)** was analyzed using PyMOL to investigate the ligand-binding environment and visualize important structural features.

2. Materials and Methods

Protein Structure Retrieval

The crystal structure of the EGFR kinase domain bound to Gefitinib was obtained from the Protein Data Bank (**PDB ID: 4WKQ**). The structure has a resolution of 1.85 Å and represents a high-quality experimental model suitable for structural analysis.

Visualization and Analysis

PyMOL was used for visualization and analysis of the protein structure. The EGFR kinase domain was displayed using cartoon representation, while the ligand Gefitinib was visualized as stick representation. Residues located within 4 Å of the ligand were identified to characterize the binding pocket.

3. Results

Overall Structure of the EGFR–Gefitinib Complex

The overall structure of EGFR revealed a typical kinase-domain architecture composed primarily of alpha helices and beta sheets. Gefitinib was observed within a cavity corresponding to the ATP-binding region of the kinase domain.

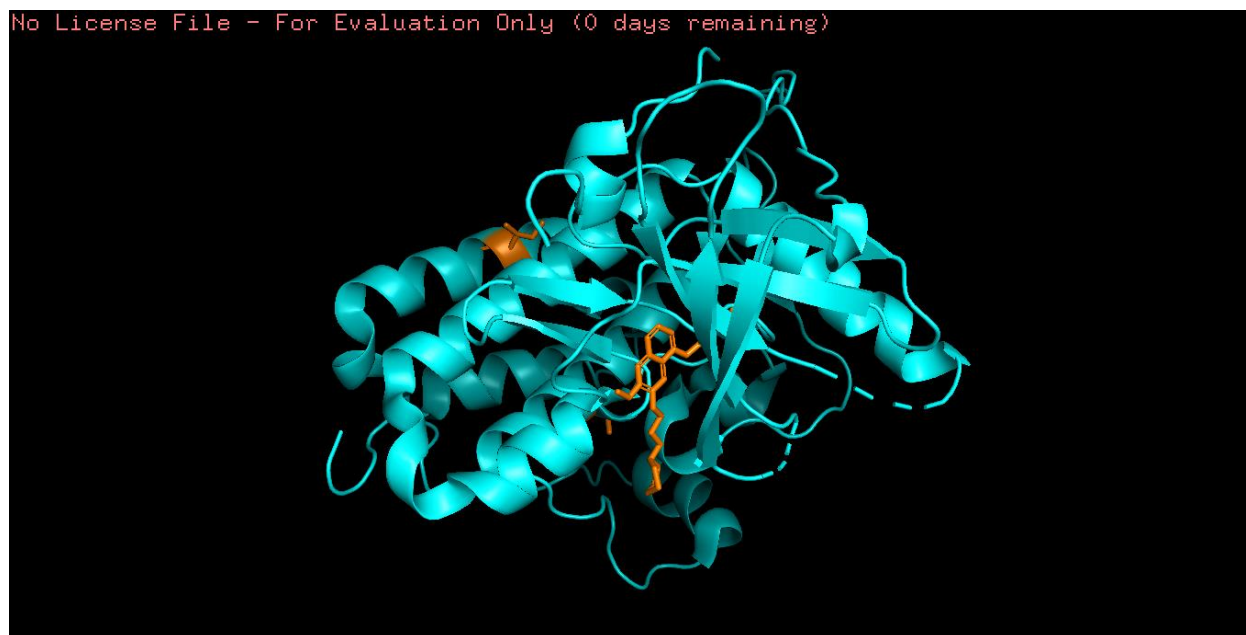


Figure 1. Overall structure of the EGFR kinase domain in complex with Gefitinib (PDB ID: 4WKQ). The protein is shown as a cyan cartoon representation, while Gefitinib is displayed as orange sticks.

Binding Pocket Analysis

Residues surrounding Gefitinib within 4 Å were identified using PyMOL. Several residues were found in close proximity to the ligand, including LEU718, LYS745, THR790, MET793, and ASP855. These residues contribute to the formation of the ligand-binding environment.

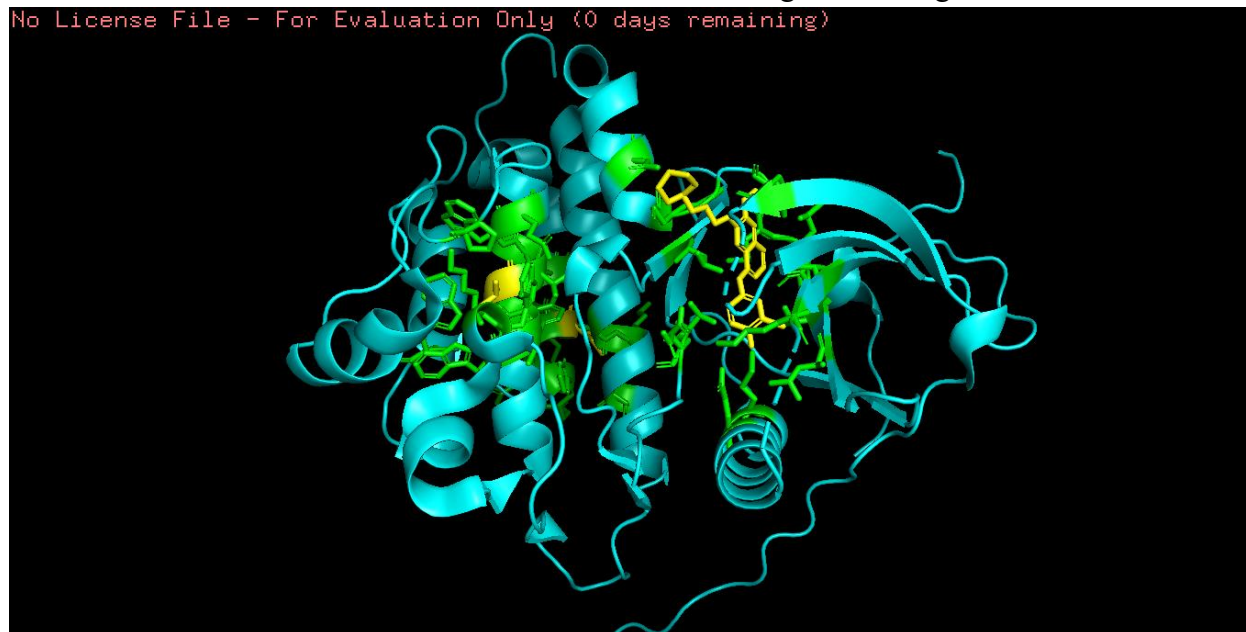


Figure 2. Binding pocket of EGFR surrounding Gefitinib. Residues within 4 Å of the ligand are shown as green sticks, highlighting the local interaction environment.

4. Conclusion

The structural analysis of the EGFR-Gefitinib complex demonstrated that Gefitinib occupies the ATP-binding pocket of the EGFR kinase domain. Visualization of the binding site revealed multiple nearby residues that contribute to ligand recognition and stabilization. This project illustrates how structural bioinformatics tools such as PyMOL can be used to investigate protein–ligand interactions and support drug discovery studies.